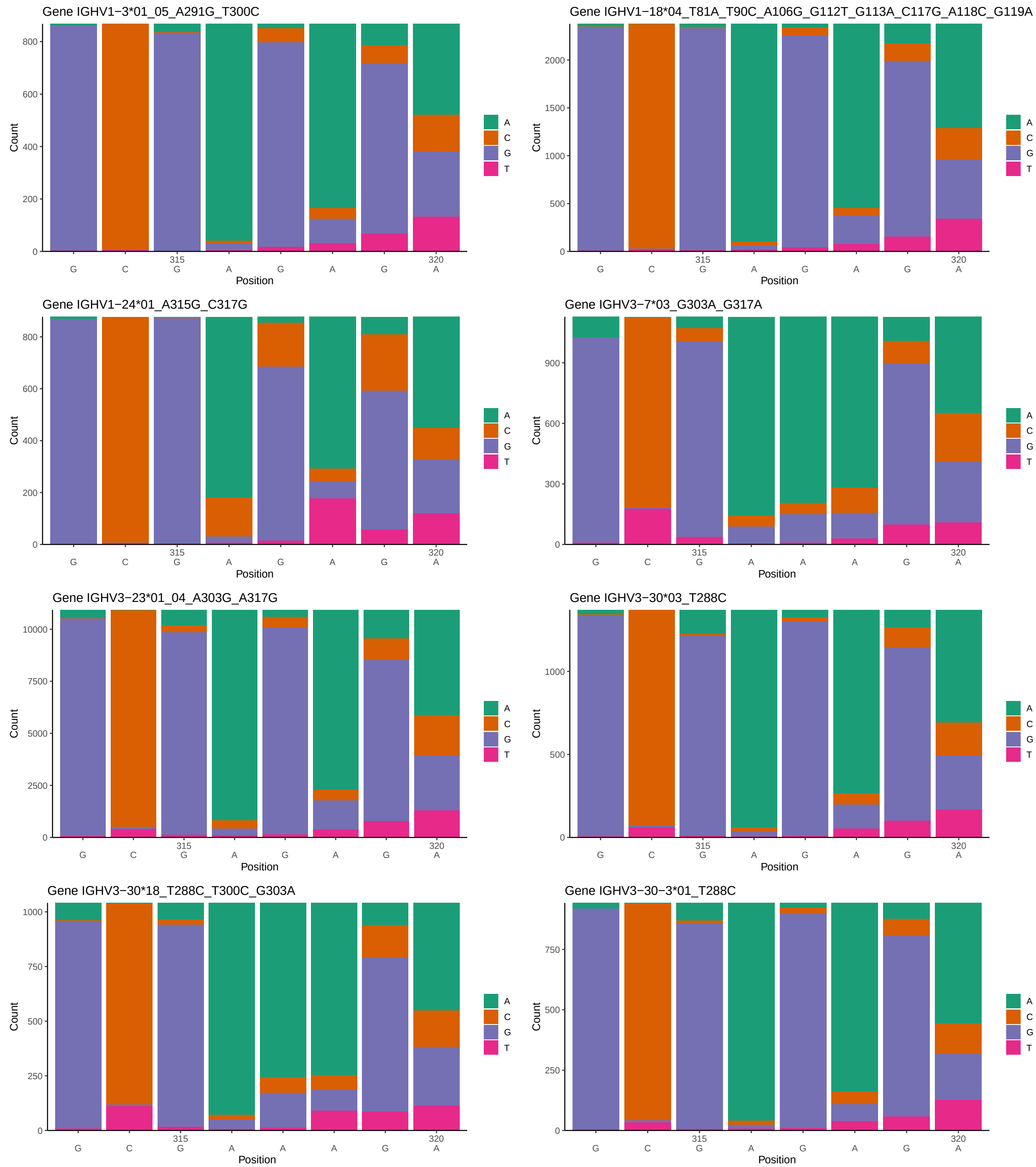
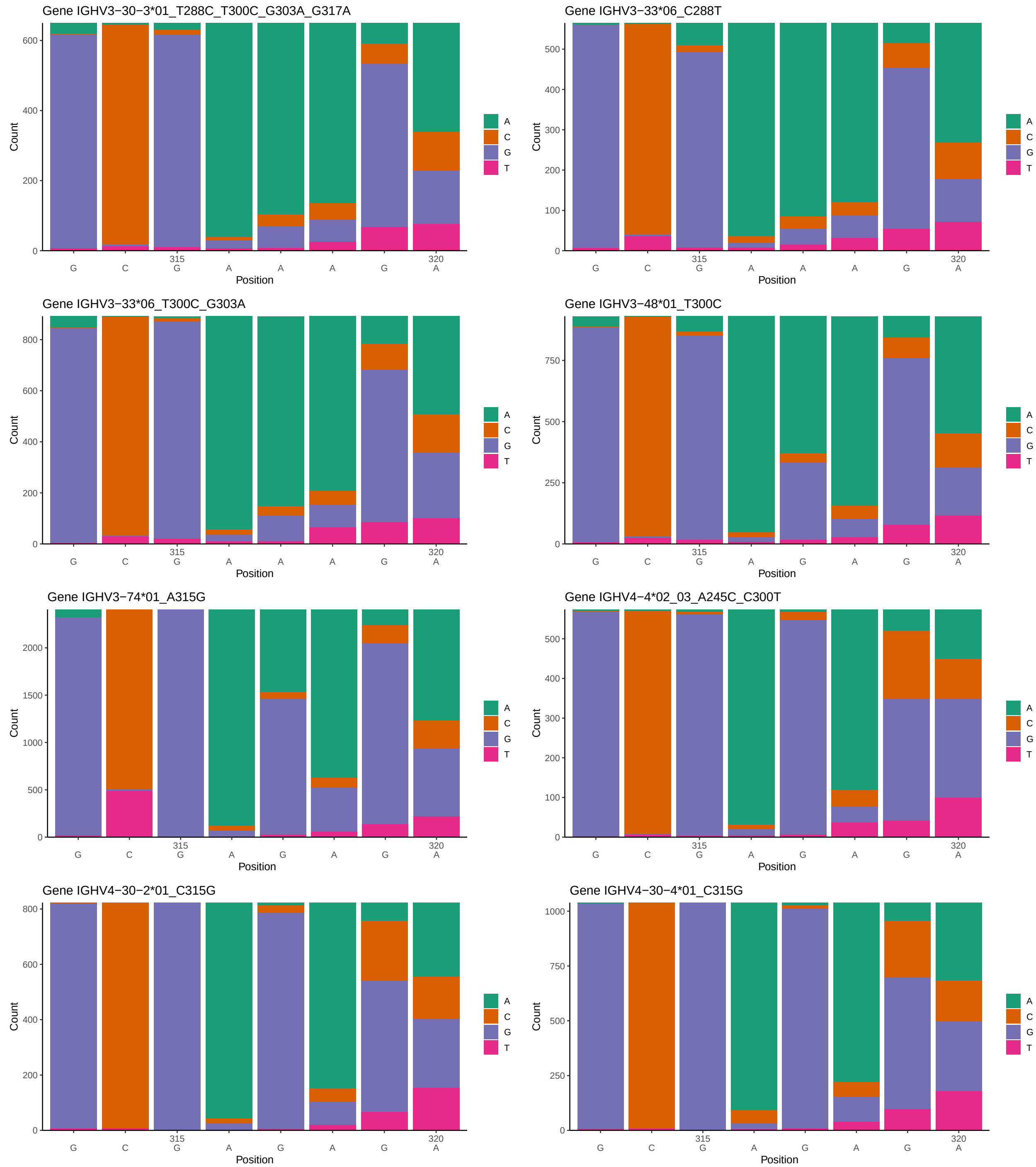
OGRDBstats Report

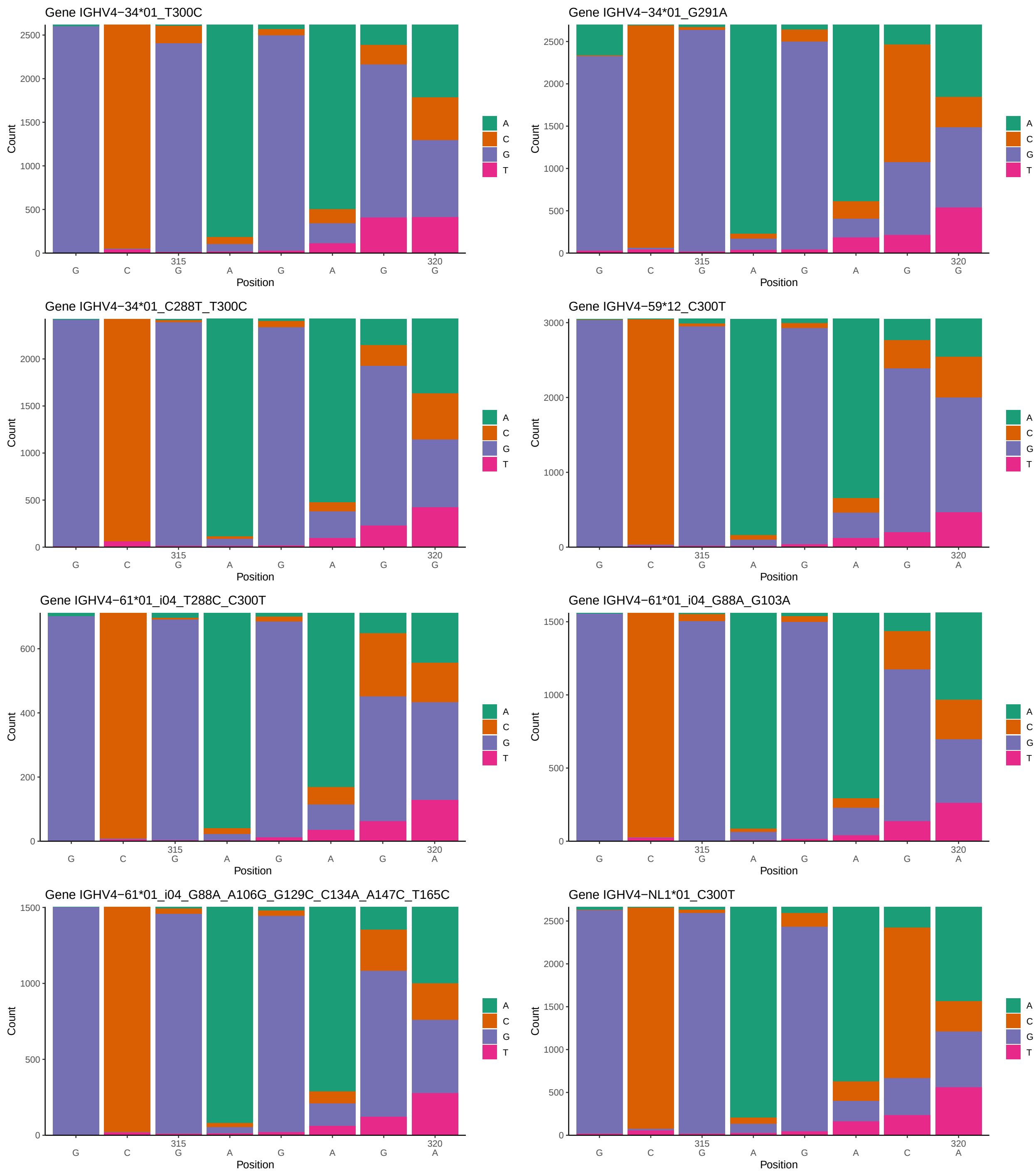
Contents

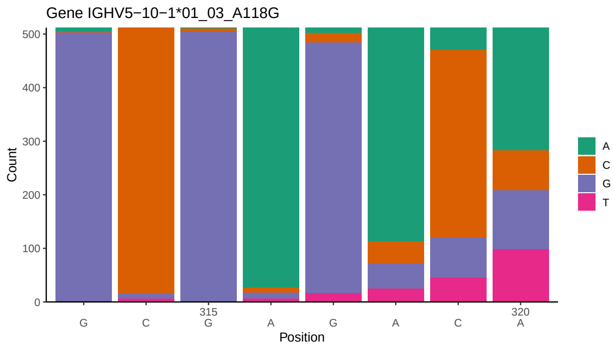
1 Novel sequence analysis

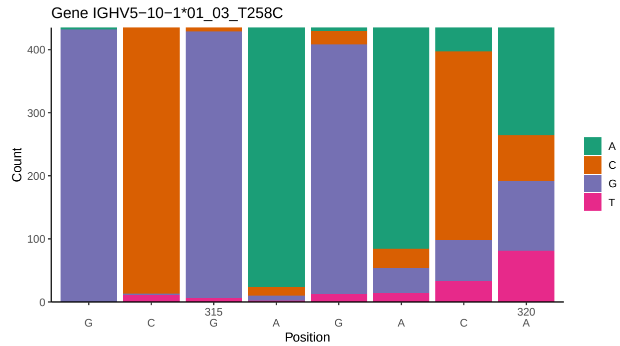
1.1 End-nucleotide composition



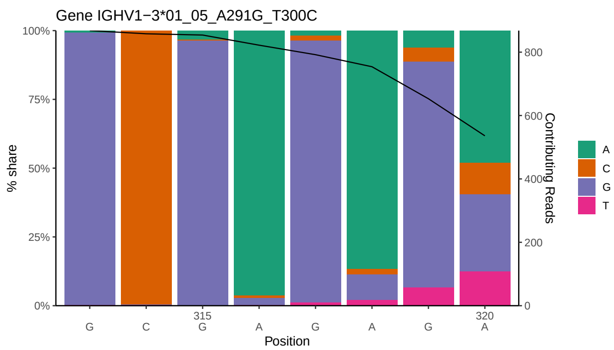

Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*01_T300C
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Count
Position
G C G A A A G A
315
320
A
C
G
T

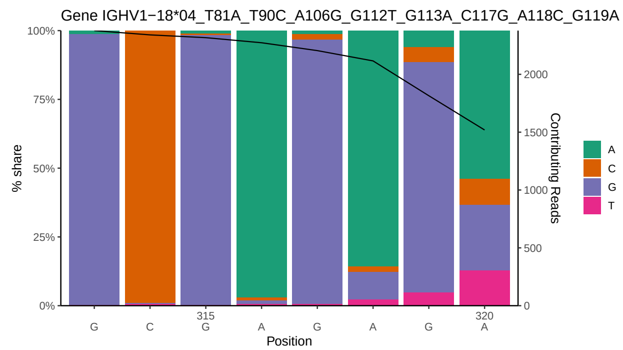

Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_G291A
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_C300T
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C
Gene IGHV4−NL1*01_C300T
Count
Position
A
C
G
T

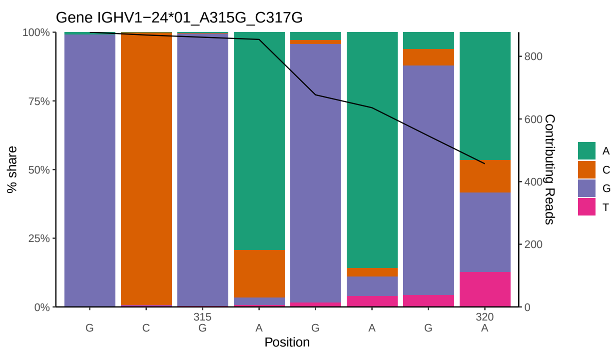


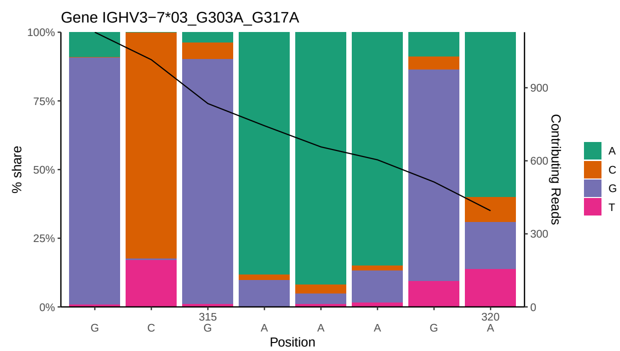


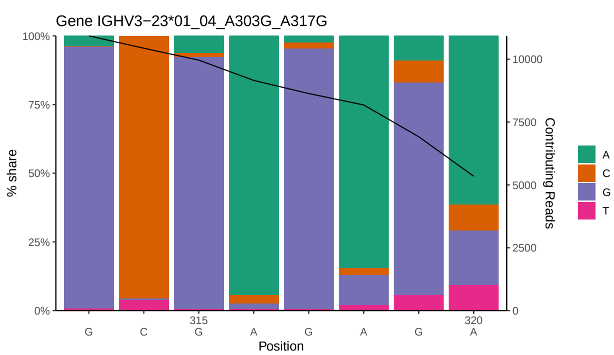
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

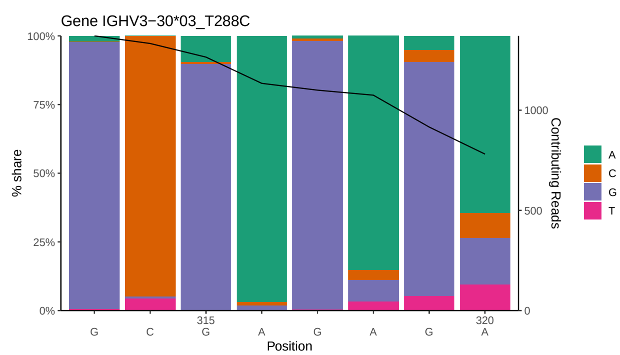


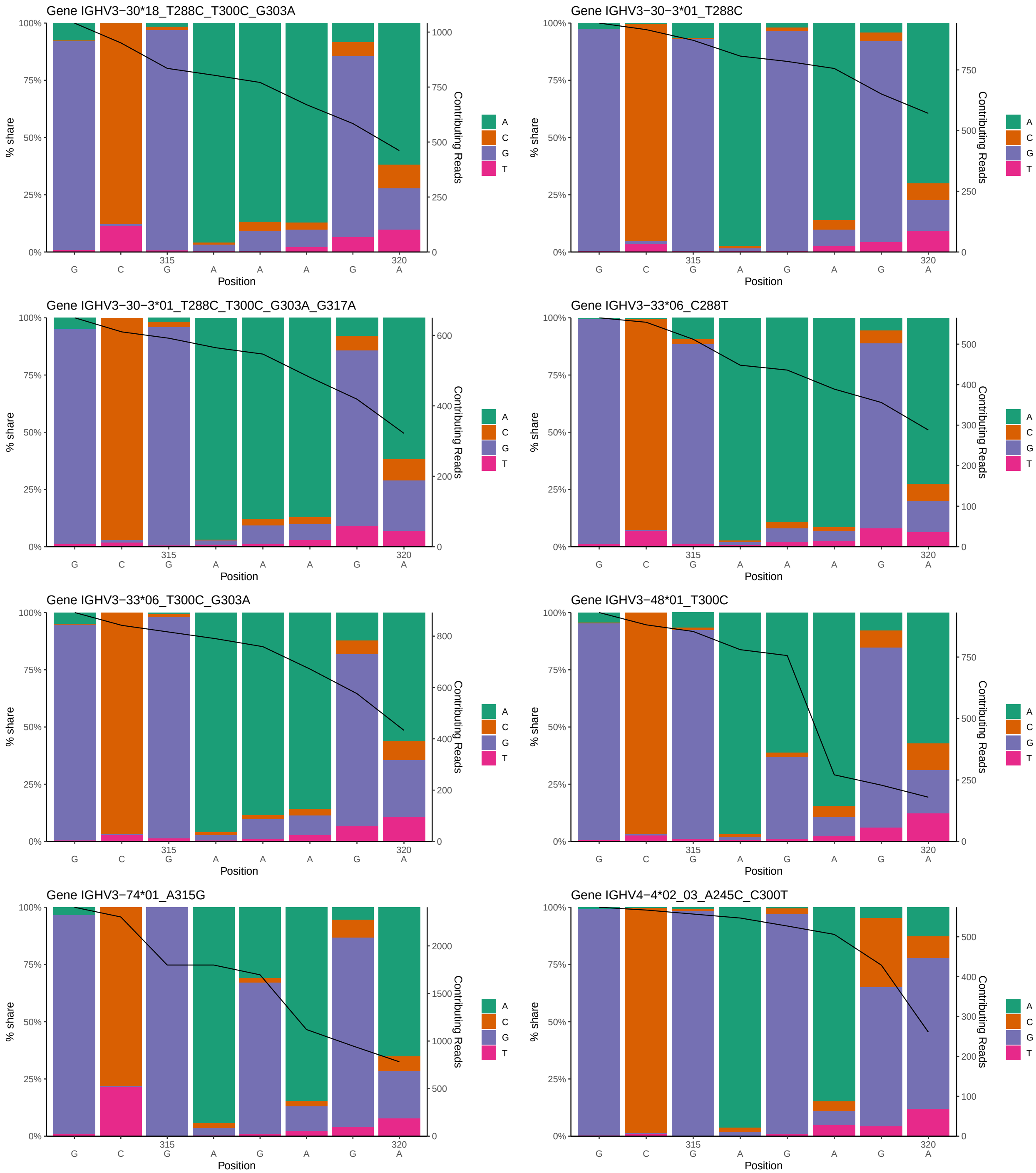


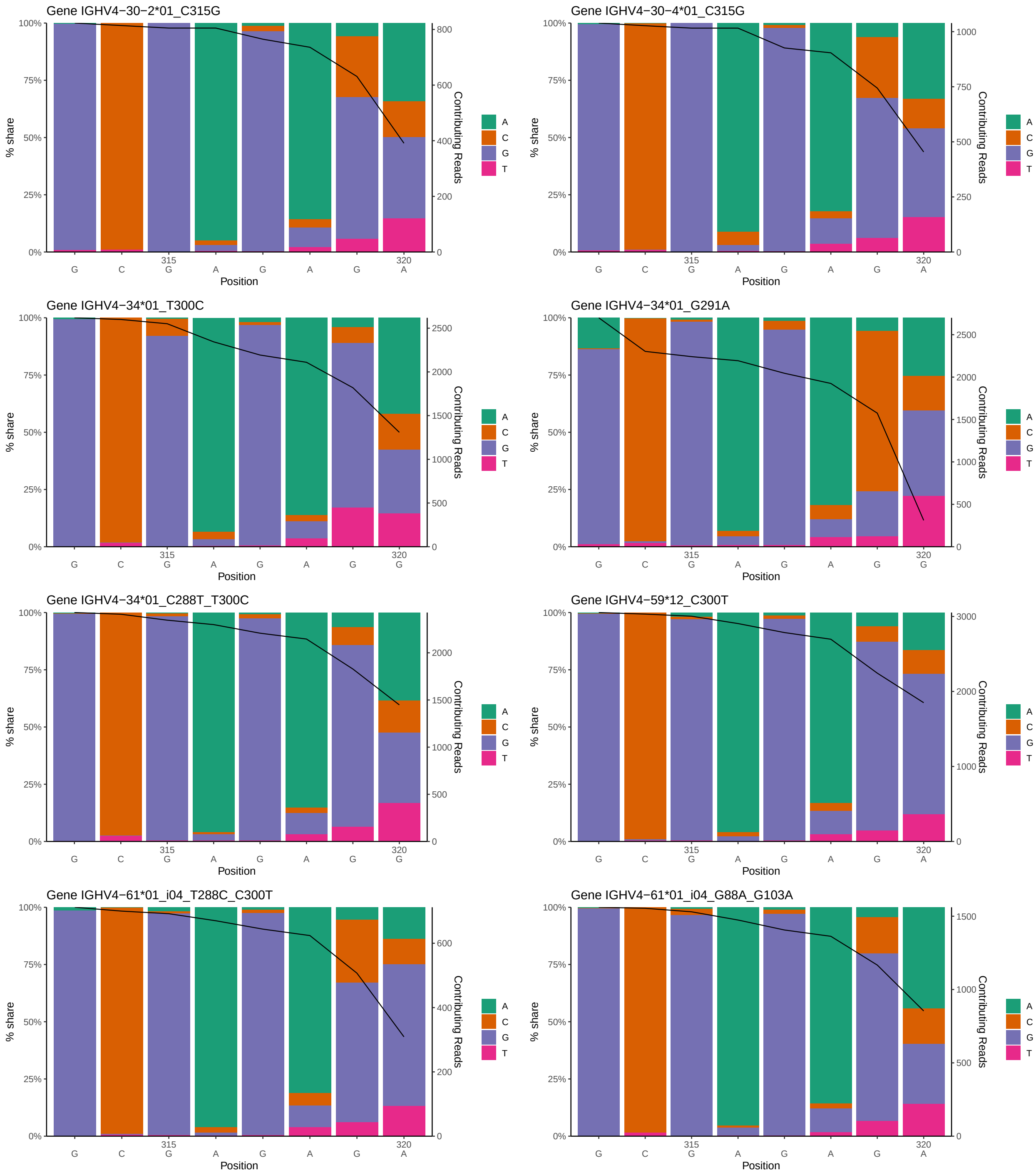


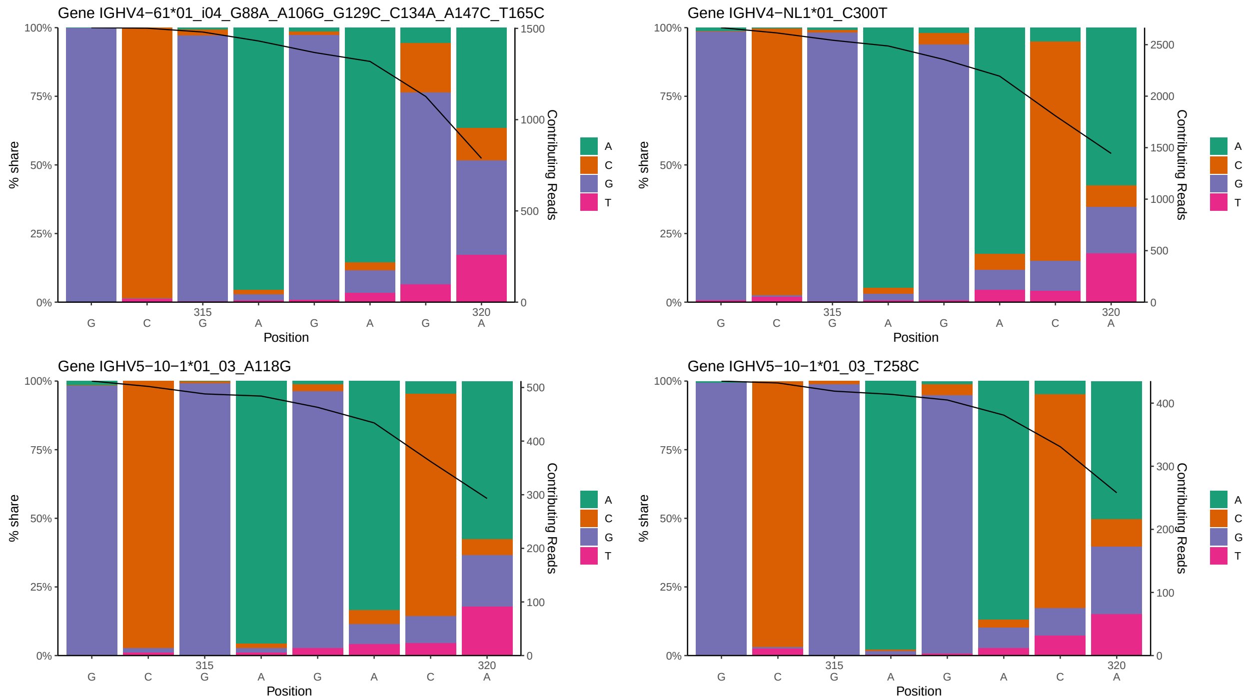




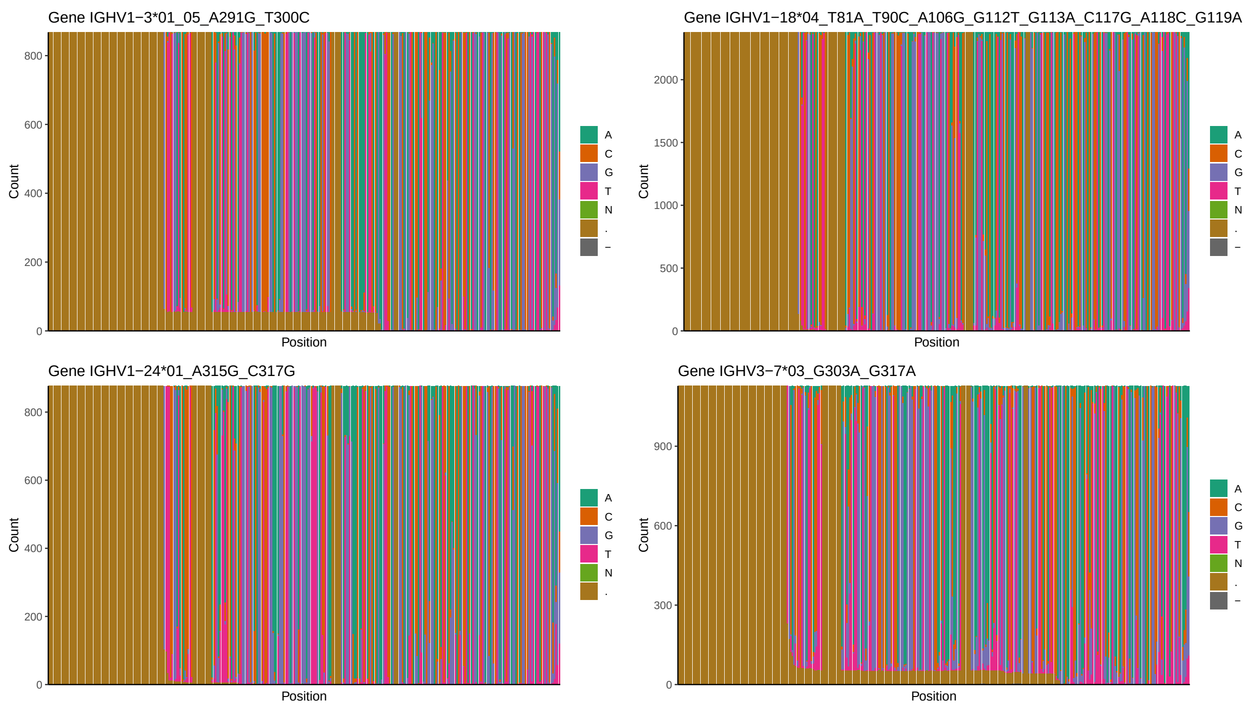


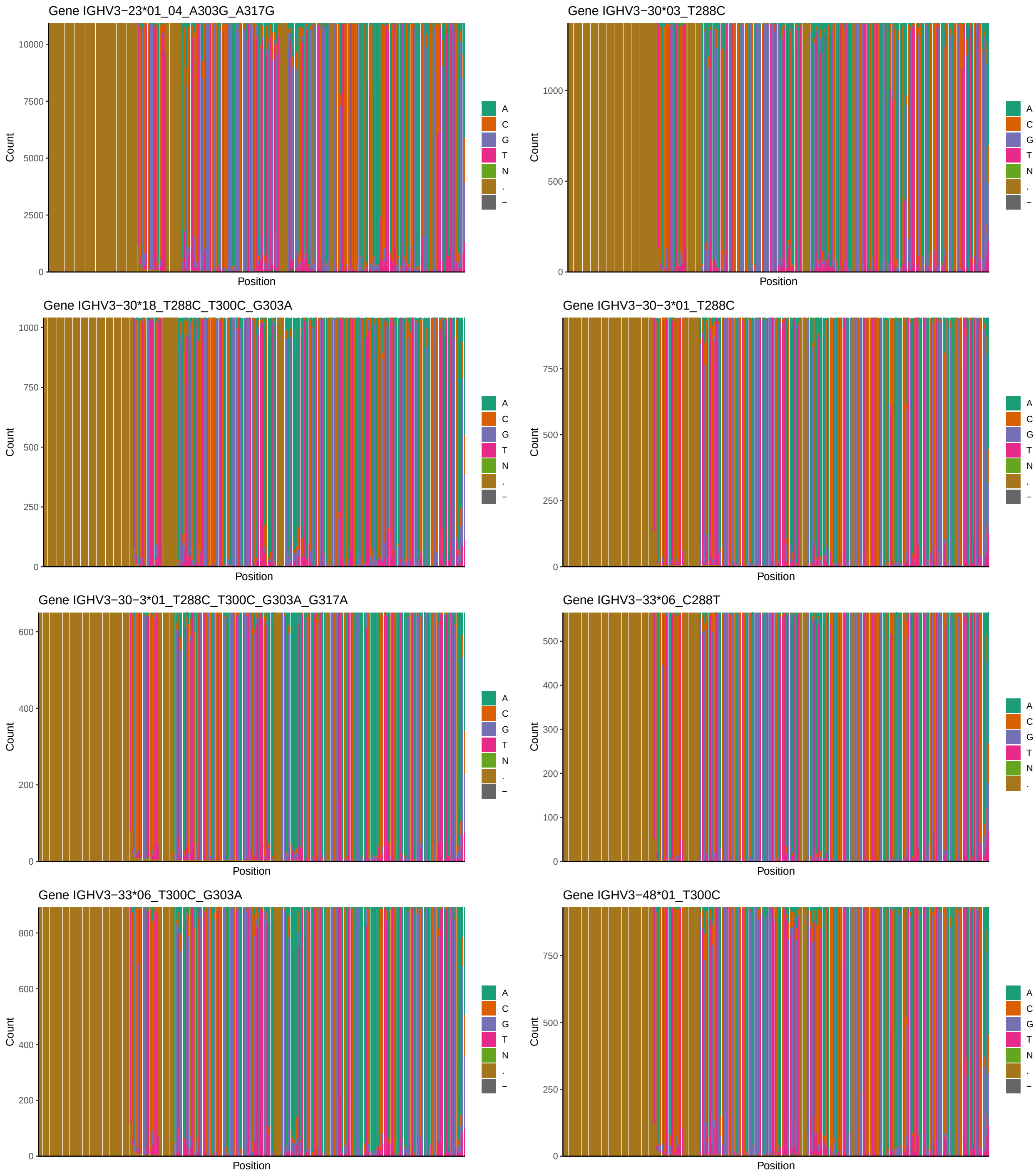

Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−30−3*01_T288C
Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*01_T300C
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
% share
Contributing Reads
Position
A
C
G
T

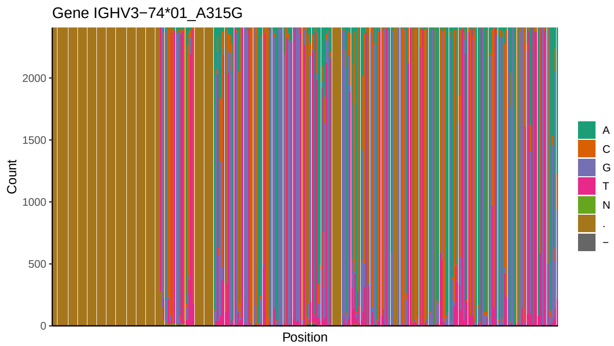

Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_G291A
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_C300T
Gene IGHV4−61*01_i04_G88A_G103A
% share
Contributing Reads
Position
100%
75%
50%
25%
0%
315
320
G C G A G A G A
G C G A G A G G
A
C
G
T

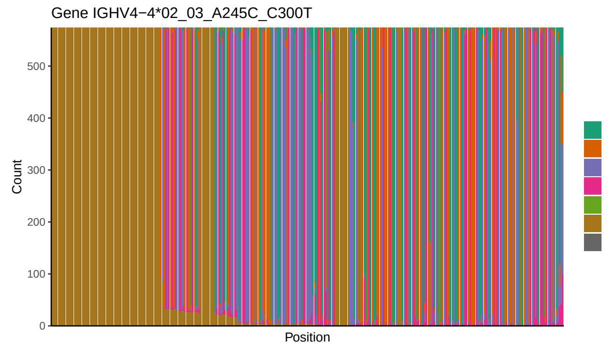


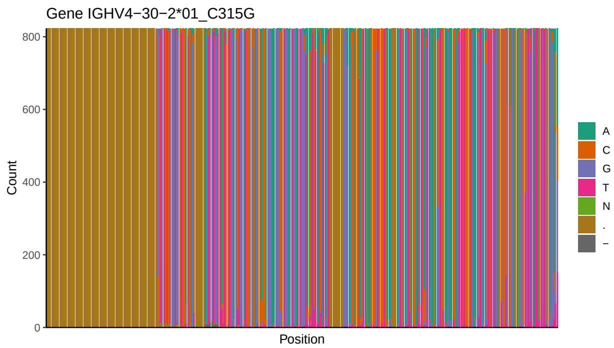
1.3 Whole-sequence composition of each assigned read

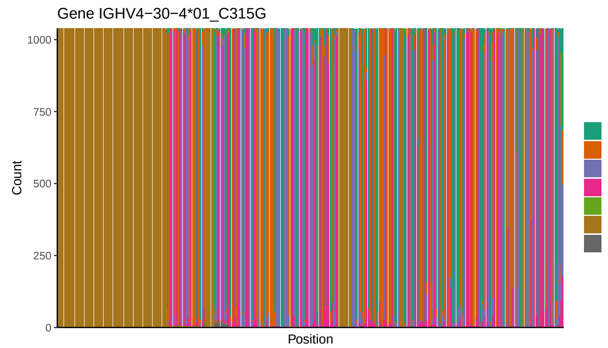


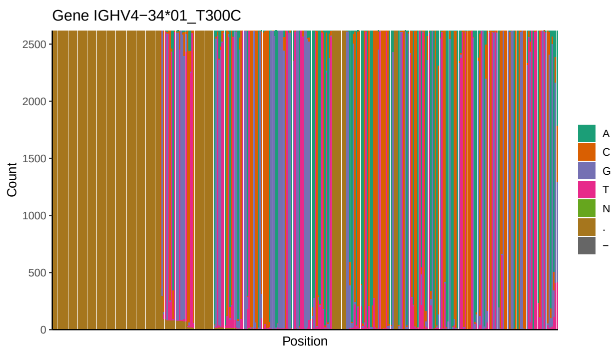

Gene IGHV3−23*01_04_A303G_A317G
Gene IGHV3−30*03_T288C
Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−30−3*01_T288C
Gene IGHV3−30−3*01_T288C_T300C_G303A_G317A
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*01_T300C
Count
Position
A
C
G
T
N
.
−

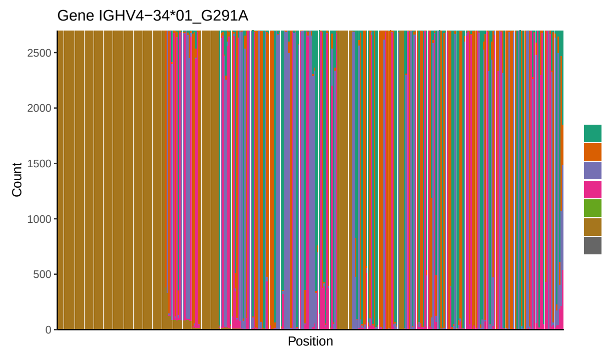

Gene IGHV3−74*01_A315G
Count
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−

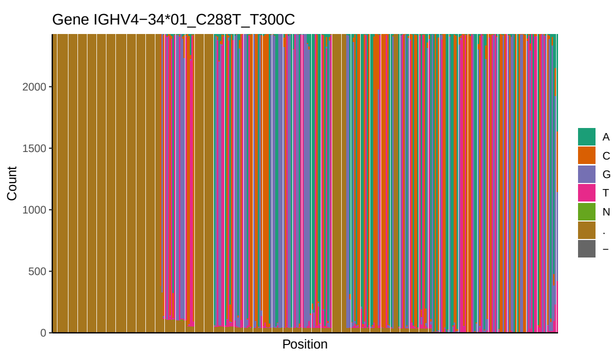

Gene IGHV4−4*02_03_A245C_C300T
Count
500
400
300
200
100
0
Position
A
C
G
T
N
.
−

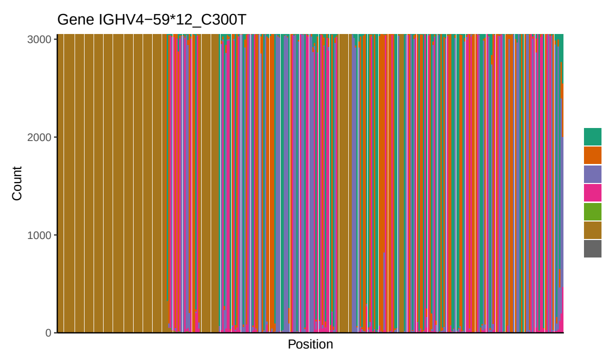

Gene IGHV4−30−2*01_C315G
Count
800
600
400
200
0
Position
A
C
G
T
N
.
−

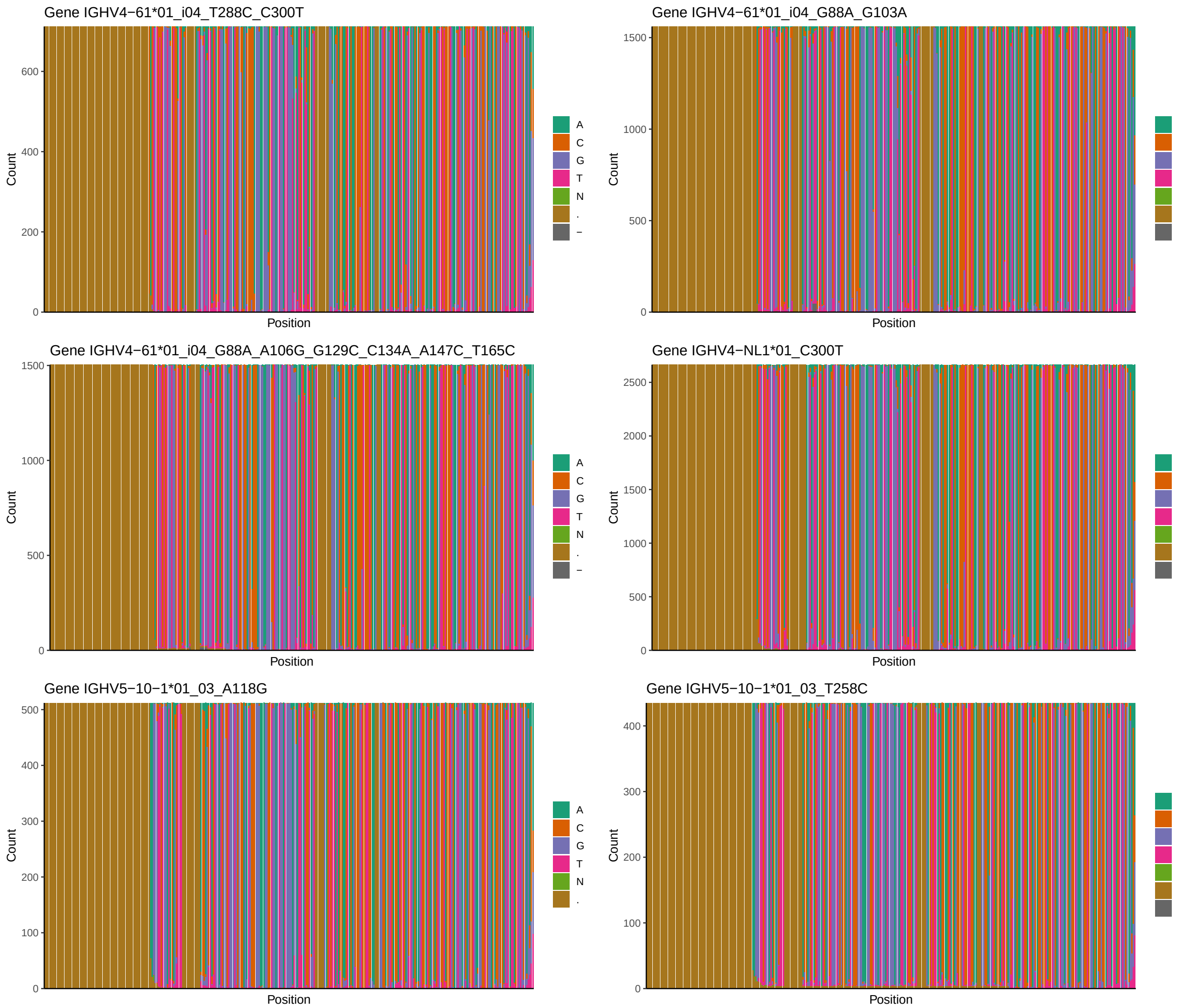

Gene IGHV4−30−4*01_C315G
Count
1000
750
500
250
0
Position
A
C
G
T
N
.
−


Gene IGHV4−34*01_T300C
Count
2500
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−

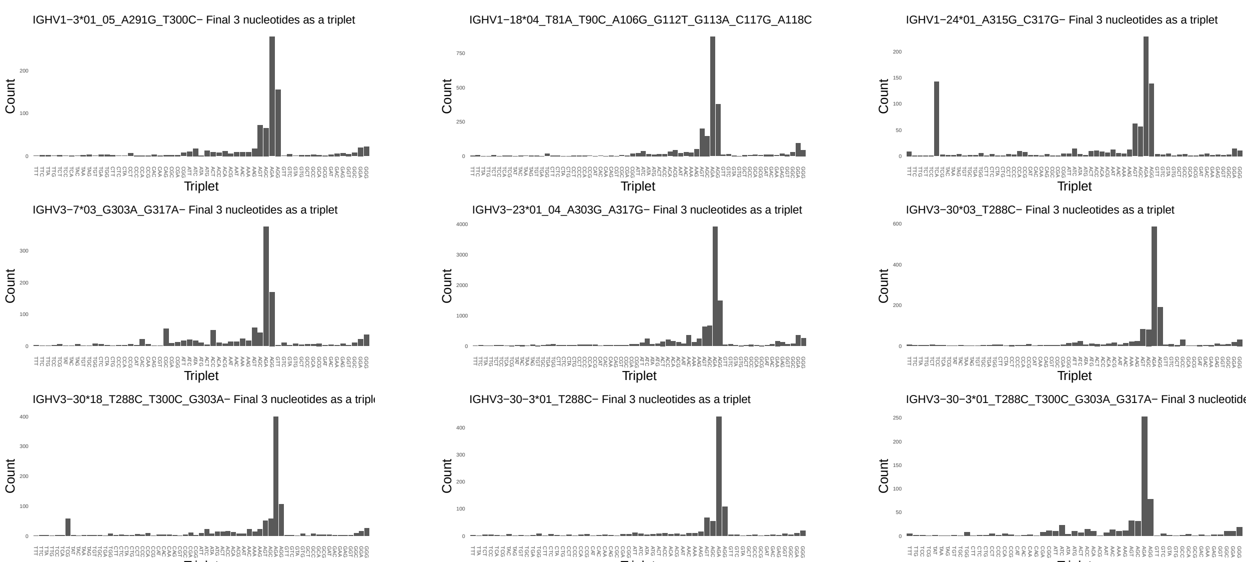

Gene IGHV4−34*01_G291A
Count
2500
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−

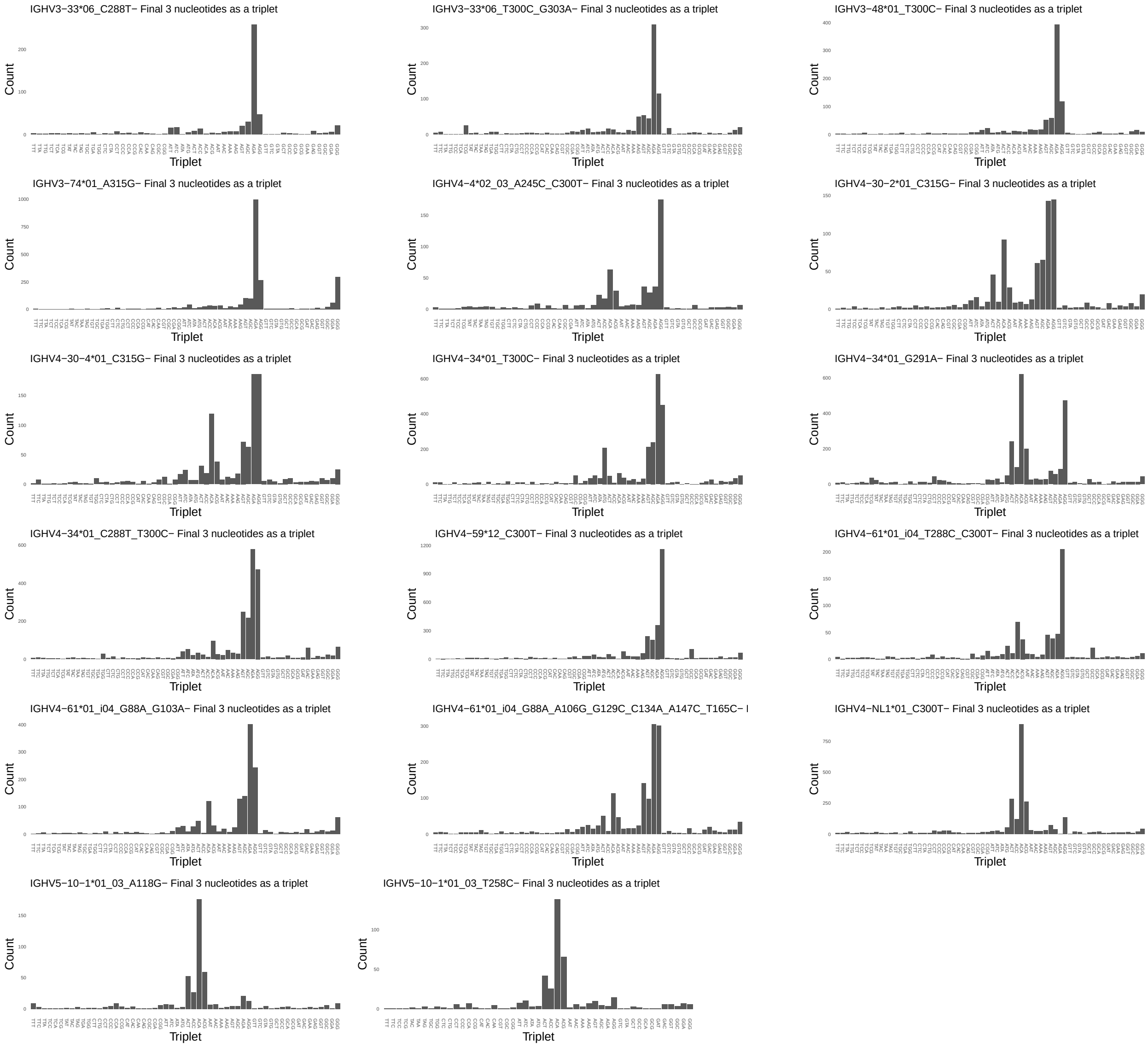

Gene IGHV4−34*01_C288T_T300C
Count
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−


Gene IGHV4−59*12_C300T
Count
3000
2000
1000
0
Position
A
C
G
T
N
.
−

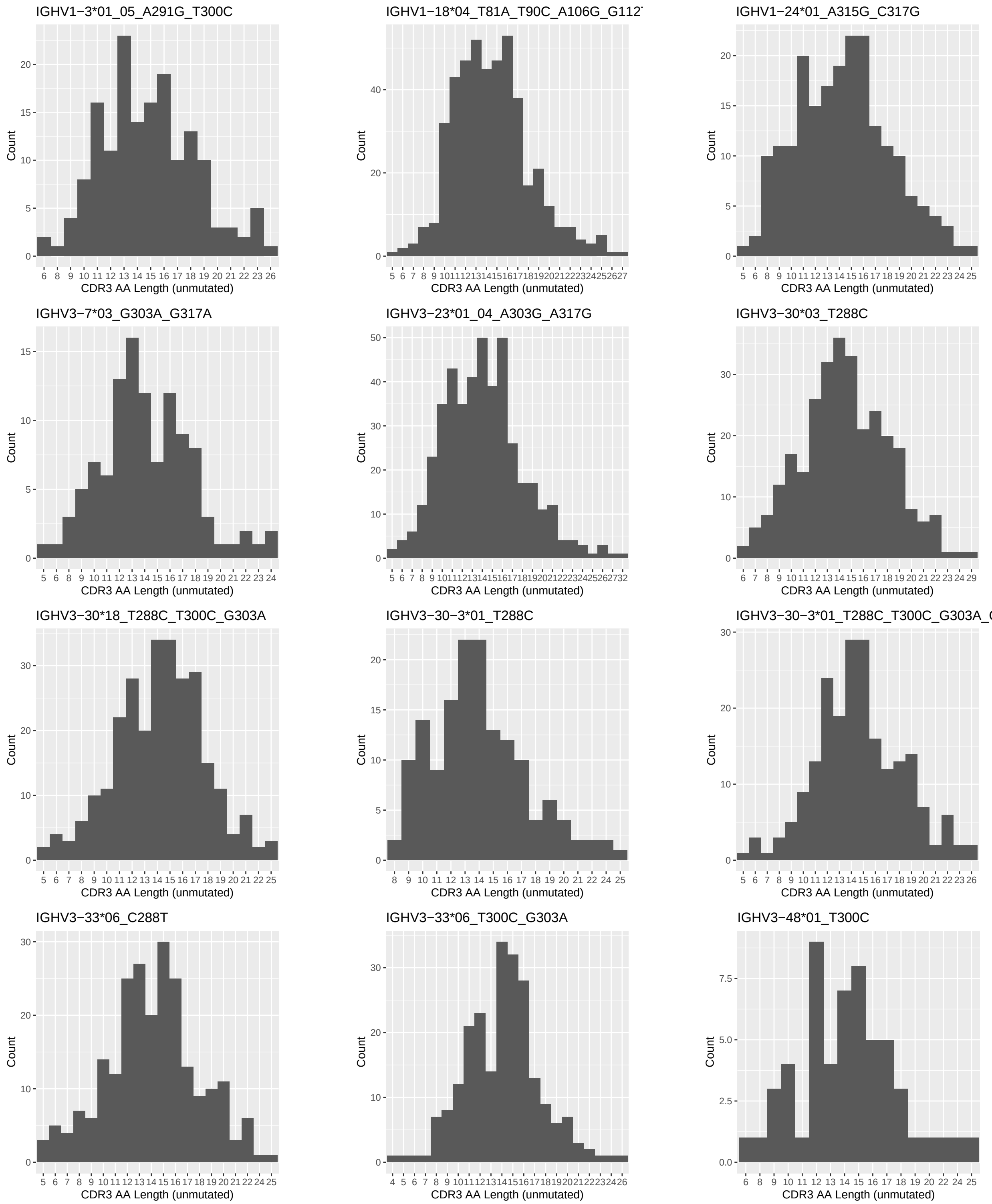


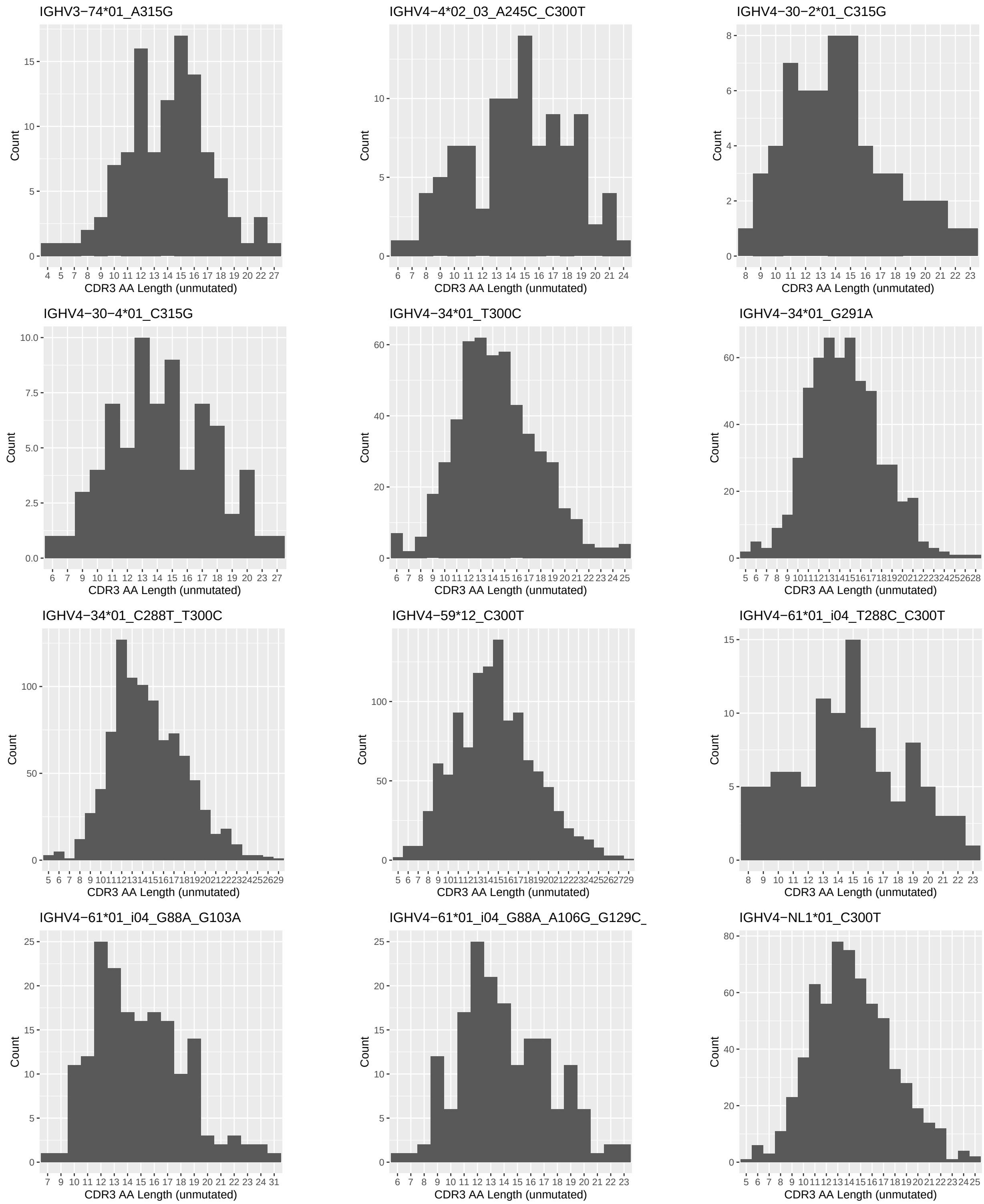
1.4 Final three nucleotides: frequency of each observed triplet

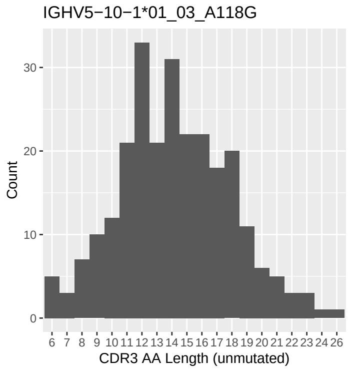


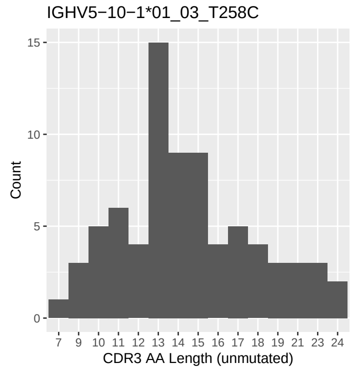

IGHV3−33*06_C288T− Final 3 nucleotides as a triplet
IGHV3−33*06_T300C_G303A− Final 3 nucleotides as a triplet
IGHV3−48*01_T300C− Final 3 nucleotides as a triplet
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_G291A− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−59*12_C300T− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_T288C_C300T− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_G103A− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C− I
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV5−10−1*01_03_A118G− Final 3 nucleotides as a triplet
IGHV5−10−1*01_03_T258C− Final 3 nucleotides as a triplet
Count
Triplet

1.5 CDR3 length distribution, in assignments to novel alleles

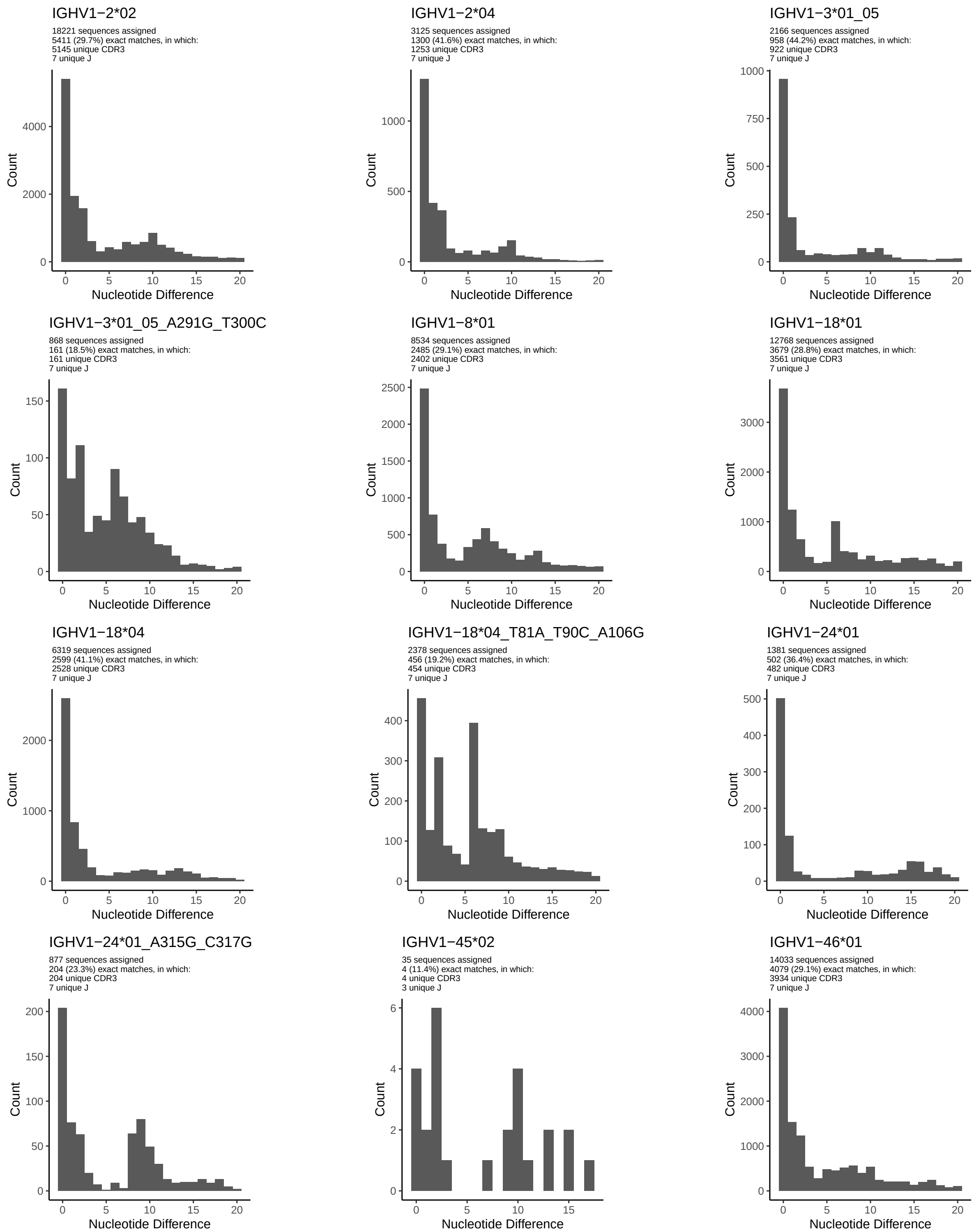


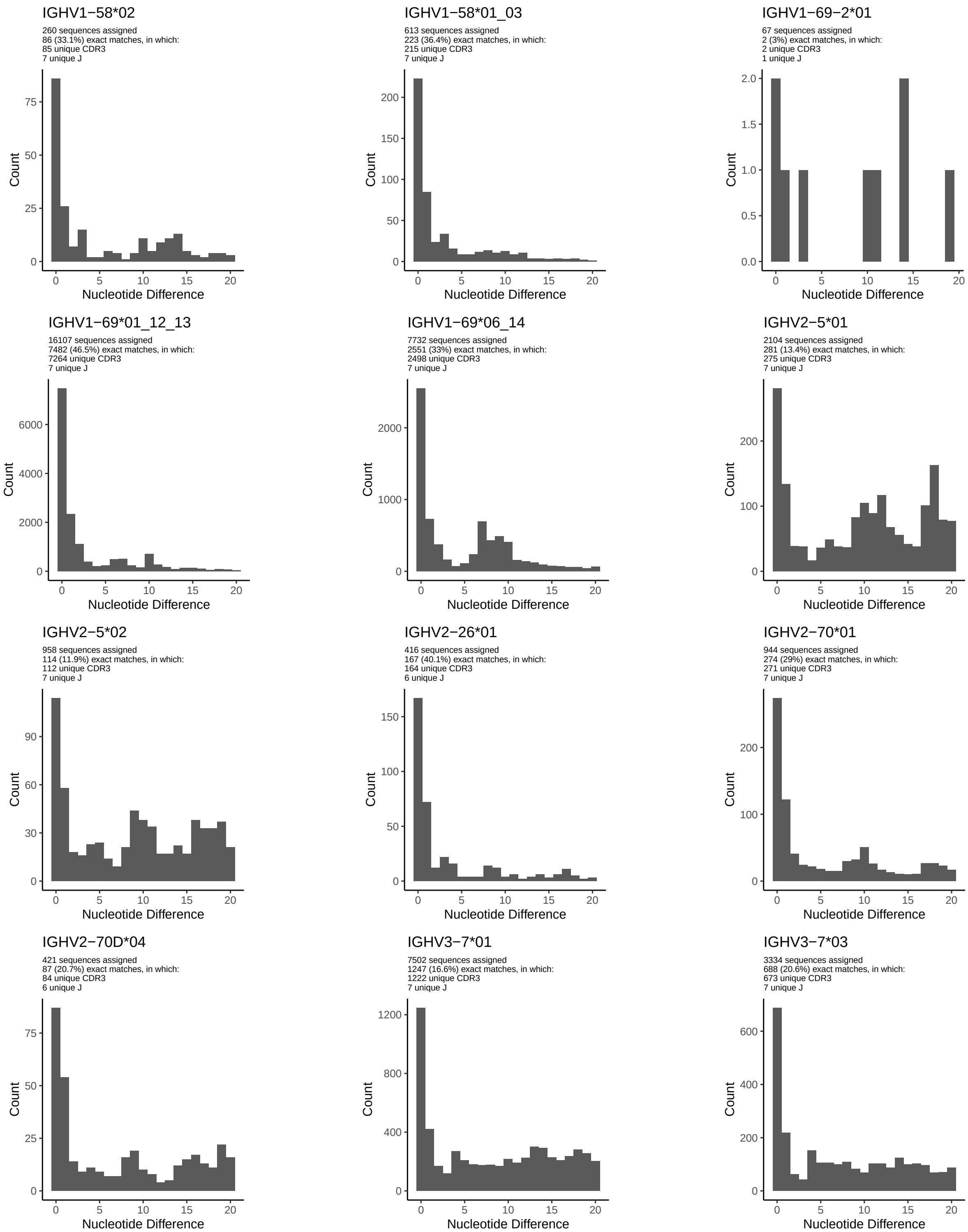

IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−4*01_C315G
IGHV4−34*01_T300C
IGHV4−34*01_G291A
IGHV4−34*01_C288T_T300C
IGHV4−59*12_C300T
IGHV4−61*01_i04_T288C_C300T
IGHV4−61*01_i04_G88A_G103A
IGHV4−61*01_i04_G88A_A106G_G129C_
IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

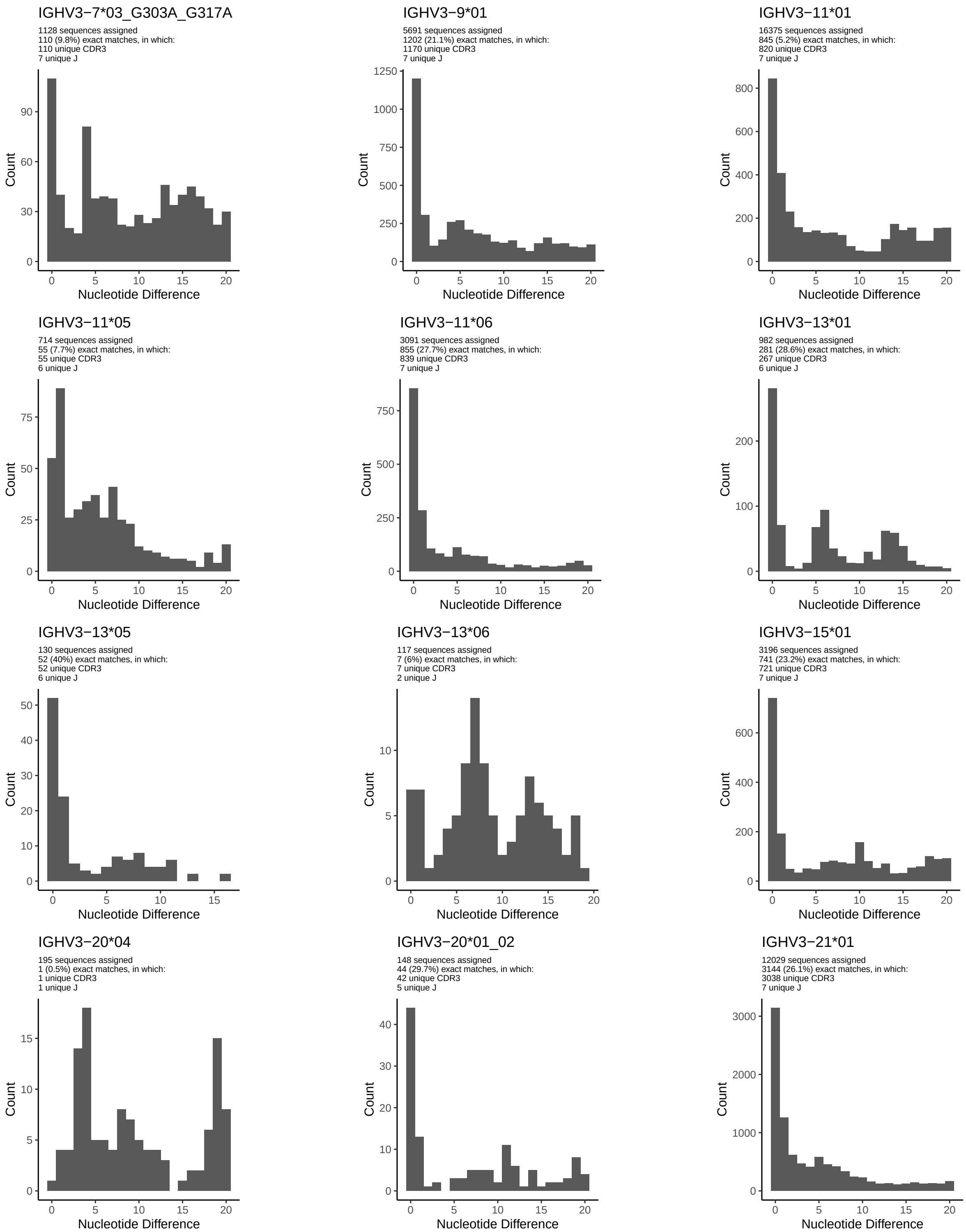

IGHV5−10−1*01_03_A118G
Count
0
10
20
30
6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26
CDR3 AA Length (unmutated)

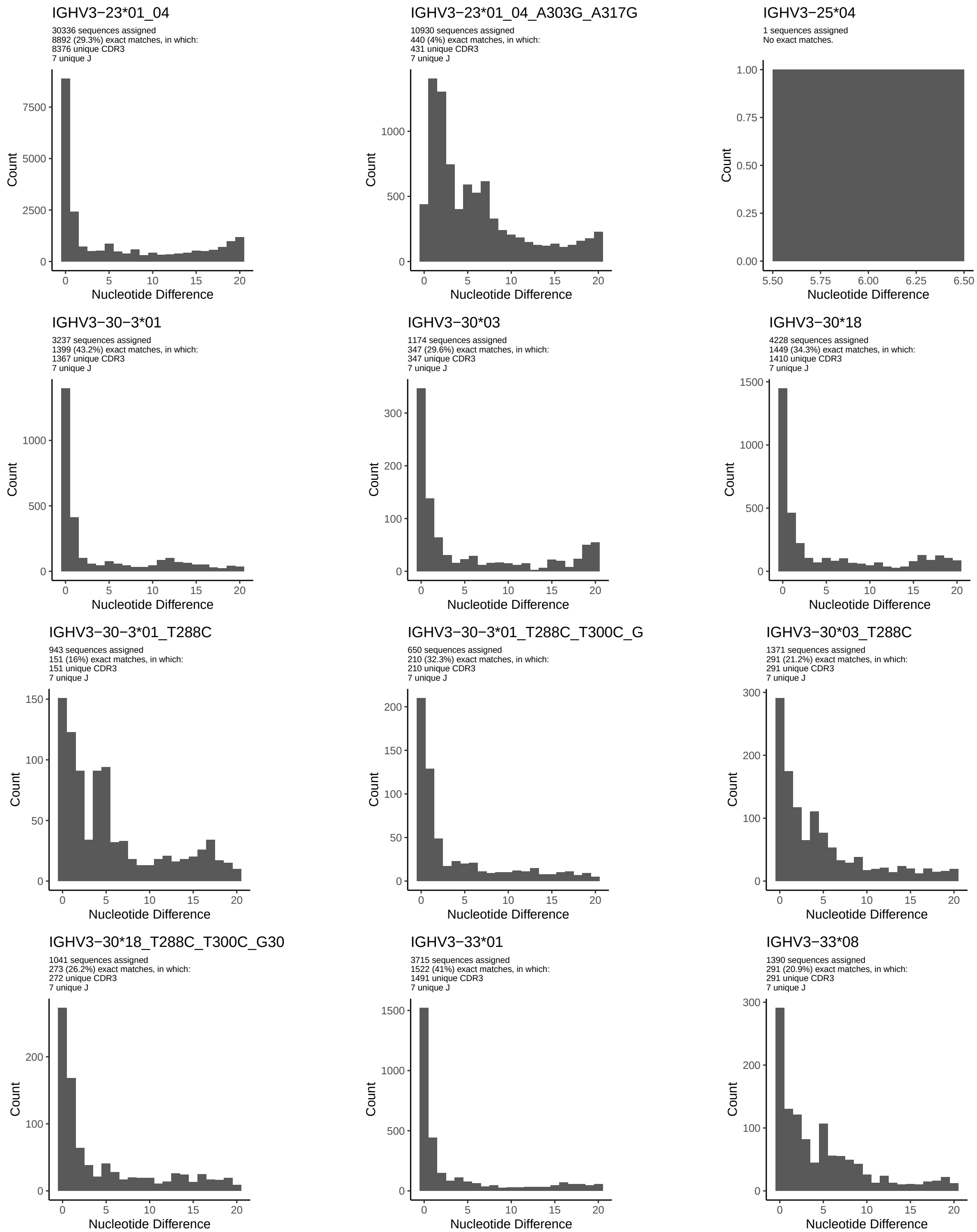

IGHV5−10−1*01_03_T258C
Count
0
5
10
15
7 9 10 11 12 13 14 15 16 17 18 19 21 23 24
CDR3 AA Length (unmutated)

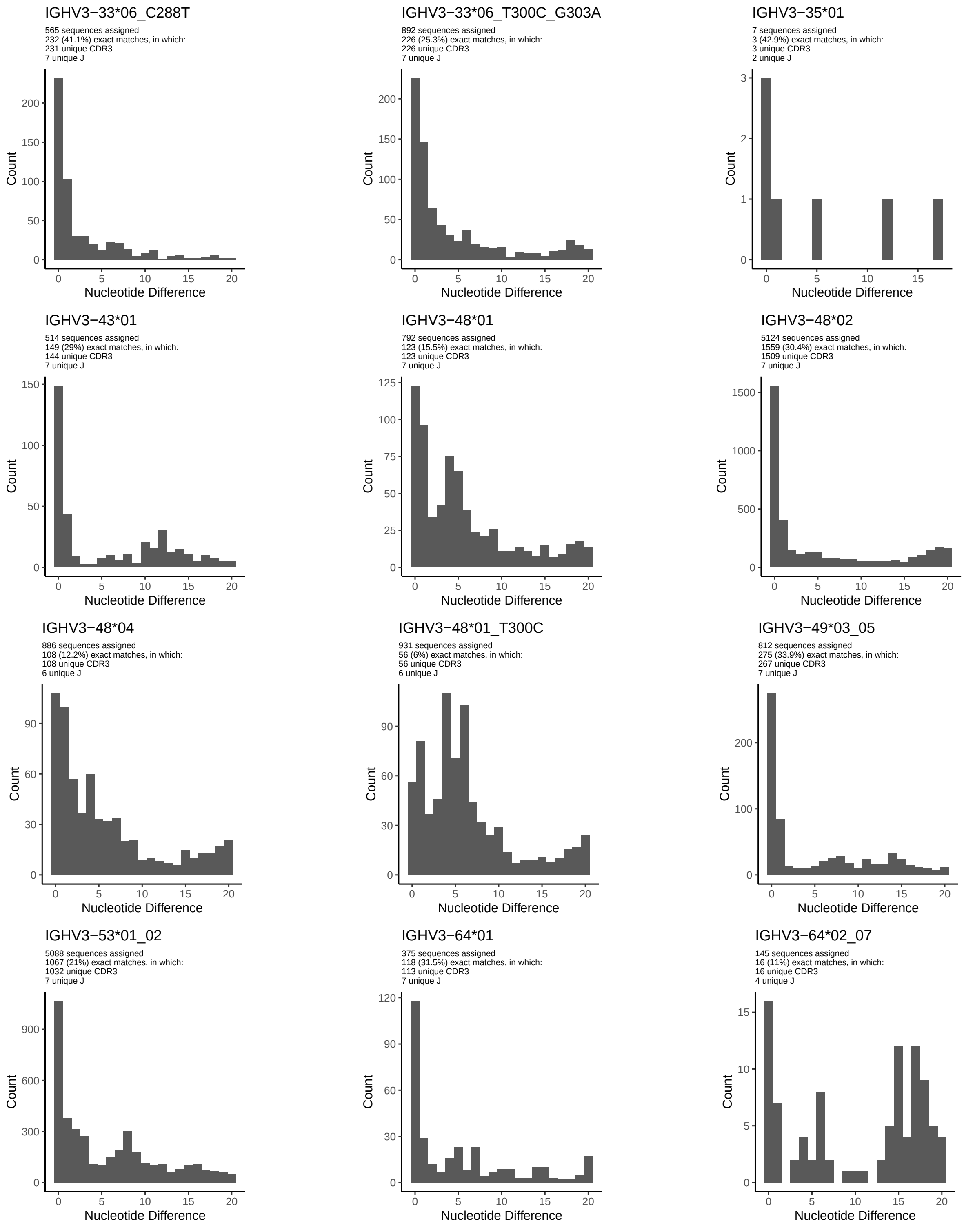
2 Variation from germline, in assignments to each allele

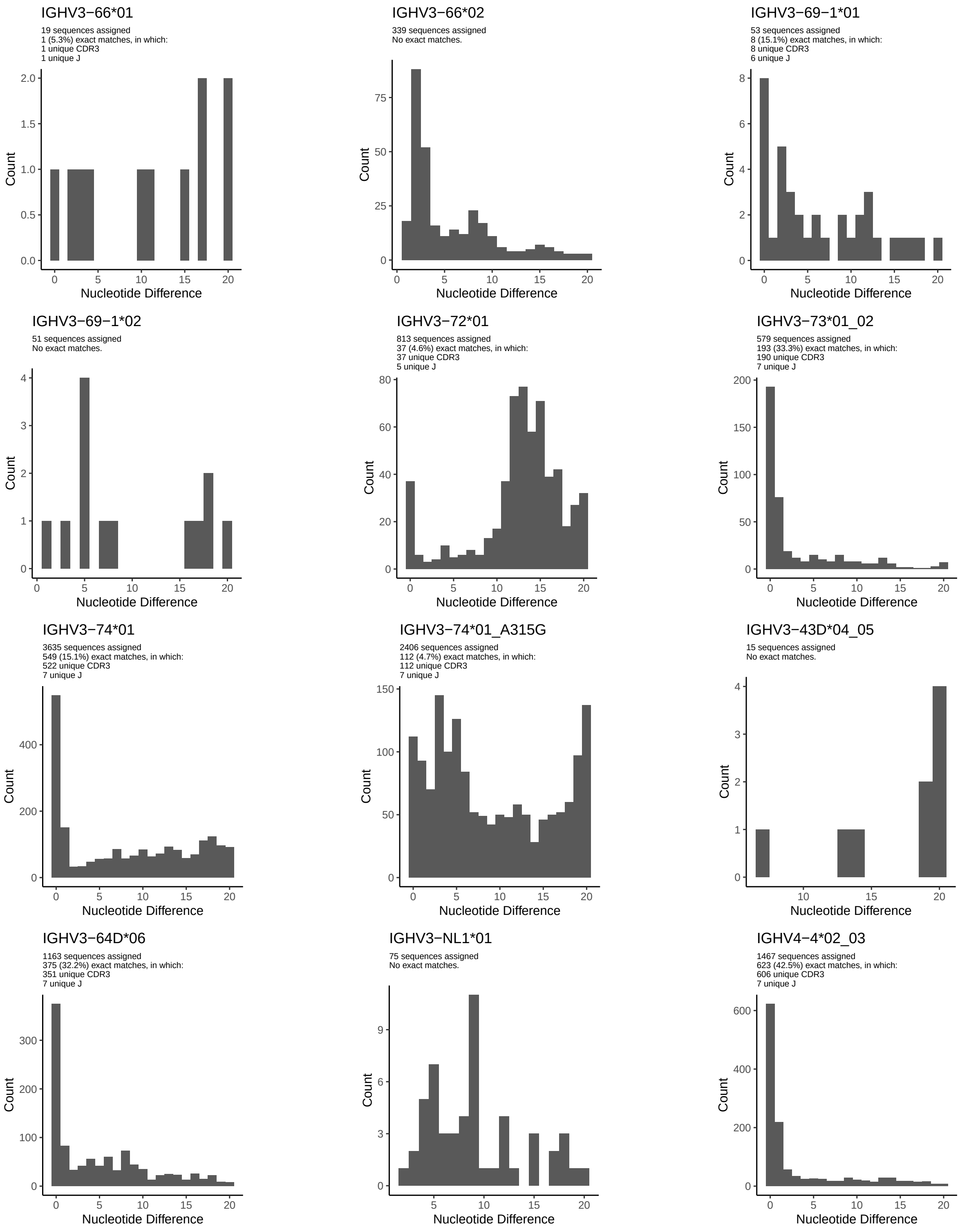


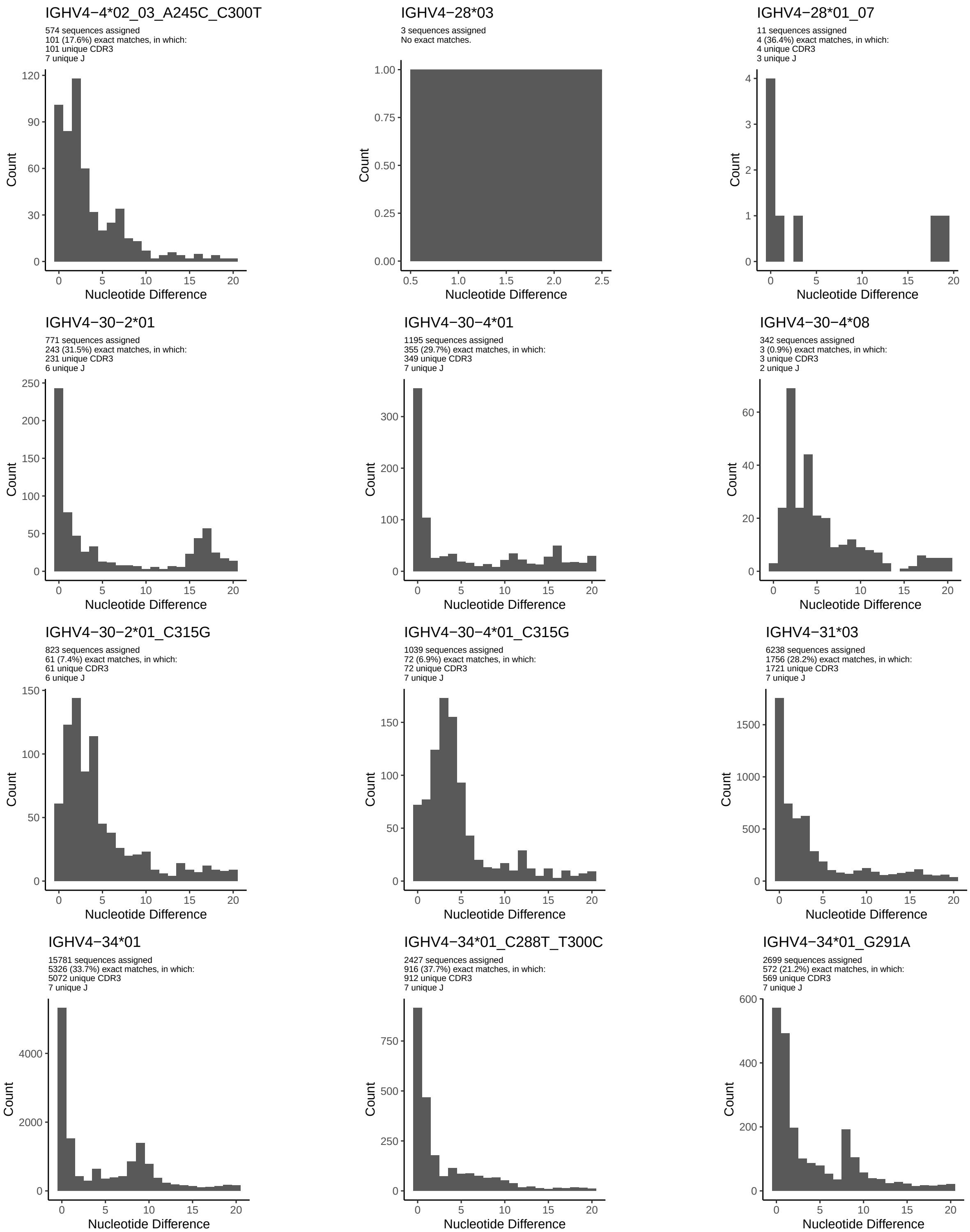

IGHV1−58*02
260 sequences assigned
86 (33.1%) exact matches, in which:
85 unique CDR3
7 unique J
IGHV1−58*01_03
613 sequences assigned
223 (36.4%) exact matches, in which:
215 unique CDR3
7 unique J
IGHV1−69−2*01
67 sequences assigned
2 (3%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV1−69*01_12_13
16107 sequences assigned
7482 (46.5%) exact matches, in which:
7264 unique CDR3
7 unique J
IGHV1−69*06_14
7732 sequences assigned
2551 (33%) exact matches, in which:
2498 unique CDR3
7 unique J
IGHV2−5*01
2104 sequences assigned
281 (13.4%) exact matches, in which:
275 unique CDR3
7 unique J
IGHV2−5*02
958 sequences assigned
114 (11.9%) exact matches, in which:
112 unique CDR3
7 unique J
IGHV2−26*01
416 sequences assigned
167 (40.1%) exact matches, in which:
164 unique CDR3
6 unique J
IGHV2−70*01
944 sequences assigned
274 (29%) exact matches, in which:
271 unique CDR3
7 unique J
IGHV2−70D*04
421 sequences assigned
87 (20.7%) exact matches, in which:
84 unique CDR3
6 unique J
IGHV3−7*01
7502 sequences assigned
1247 (16.6%) exact matches, in which:
1222 unique CDR3
7 unique J
IGHV3−7*03
3334 sequences assigned
688 (20.6%) exact matches, in which:
673 unique CDR3
7 unique J
Count
Nucleotide Difference

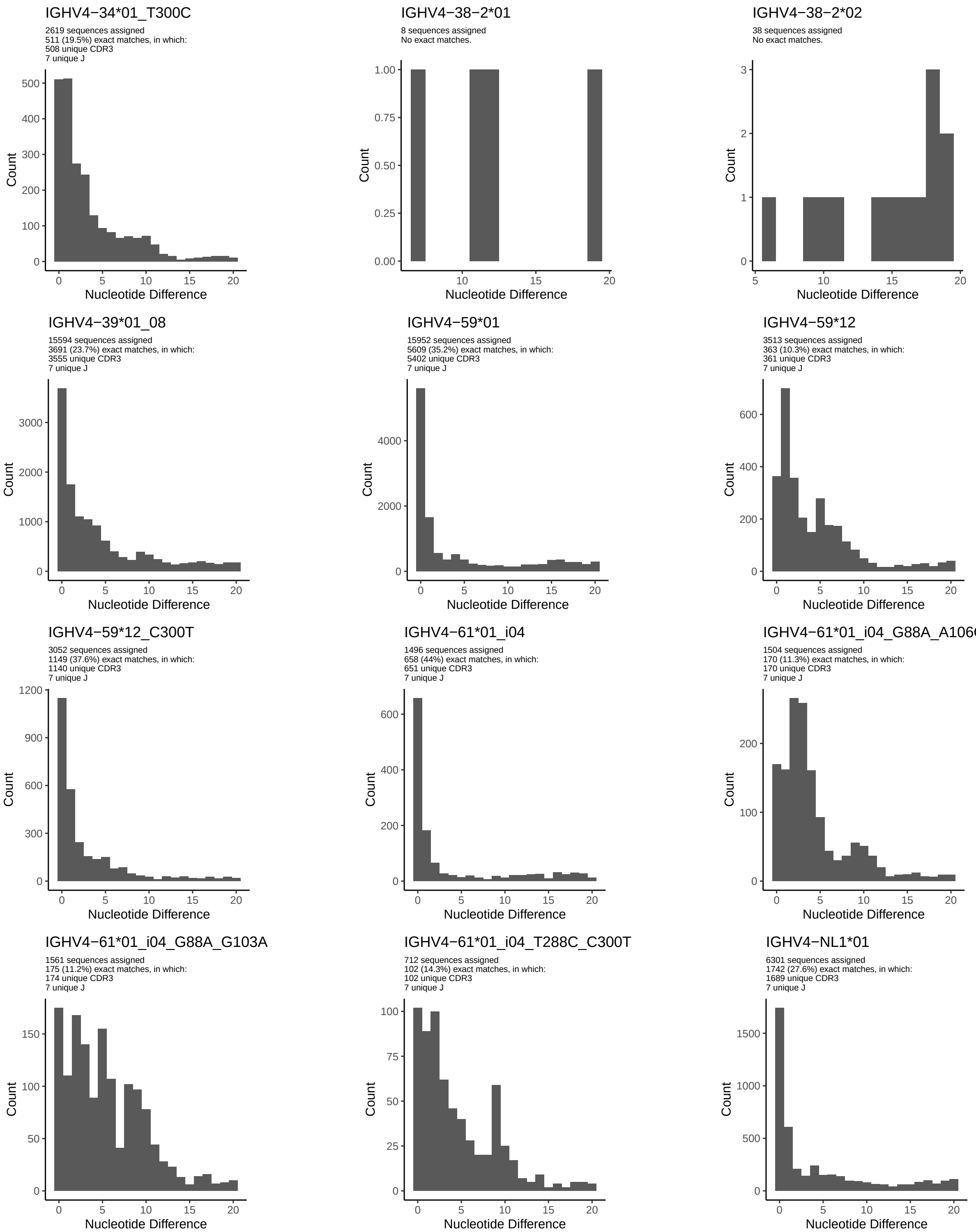

IGHV3−7*03_G303A_G317A
1128 sequences assigned
110 (9.8%) exact matches, in which:
110 unique CDR3
7 unique J
IGHV3−9*01
5691 sequences assigned
1202 (21.1%) exact matches, in which:
1170 unique CDR3
7 unique J
IGHV3−11*01
16375 sequences assigned
845 (5.2%) exact matches, in which:
820 unique CDR3
7 unique J
IGHV3−11*05
714 sequences assigned
55 (7.7%) exact matches, in which:
55 unique CDR3
6 unique J
IGHV3−11*06
3091 sequences assigned
855 (27.7%) exact matches, in which:
839 unique CDR3
7 unique J
IGHV3−13*01
982 sequences assigned
281 (28.6%) exact matches, in which:
267 unique CDR3
6 unique J
IGHV3−13*05
130 sequences assigned
52 (40%) exact matches, in which:
52 unique CDR3
6 unique J
IGHV3−13*06
117 sequences assigned
7 (6%) exact matches, in which:
7 unique CDR3
2 unique J
IGHV3−15*01
3196 sequences assigned
741 (23.2%) exact matches, in which:
721 unique CDR3
7 unique J
IGHV3−20*04
195 sequences assigned
1 (0.5%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−20*01_02
148 sequences assigned
44 (29.7%) exact matches, in which:
42 unique CDR3
5 unique J
IGHV3−21*01
12029 sequences assigned
3144 (26.1%) exact matches, in which:
3038 unique CDR3
7 unique J
Count
Nucleotide Difference

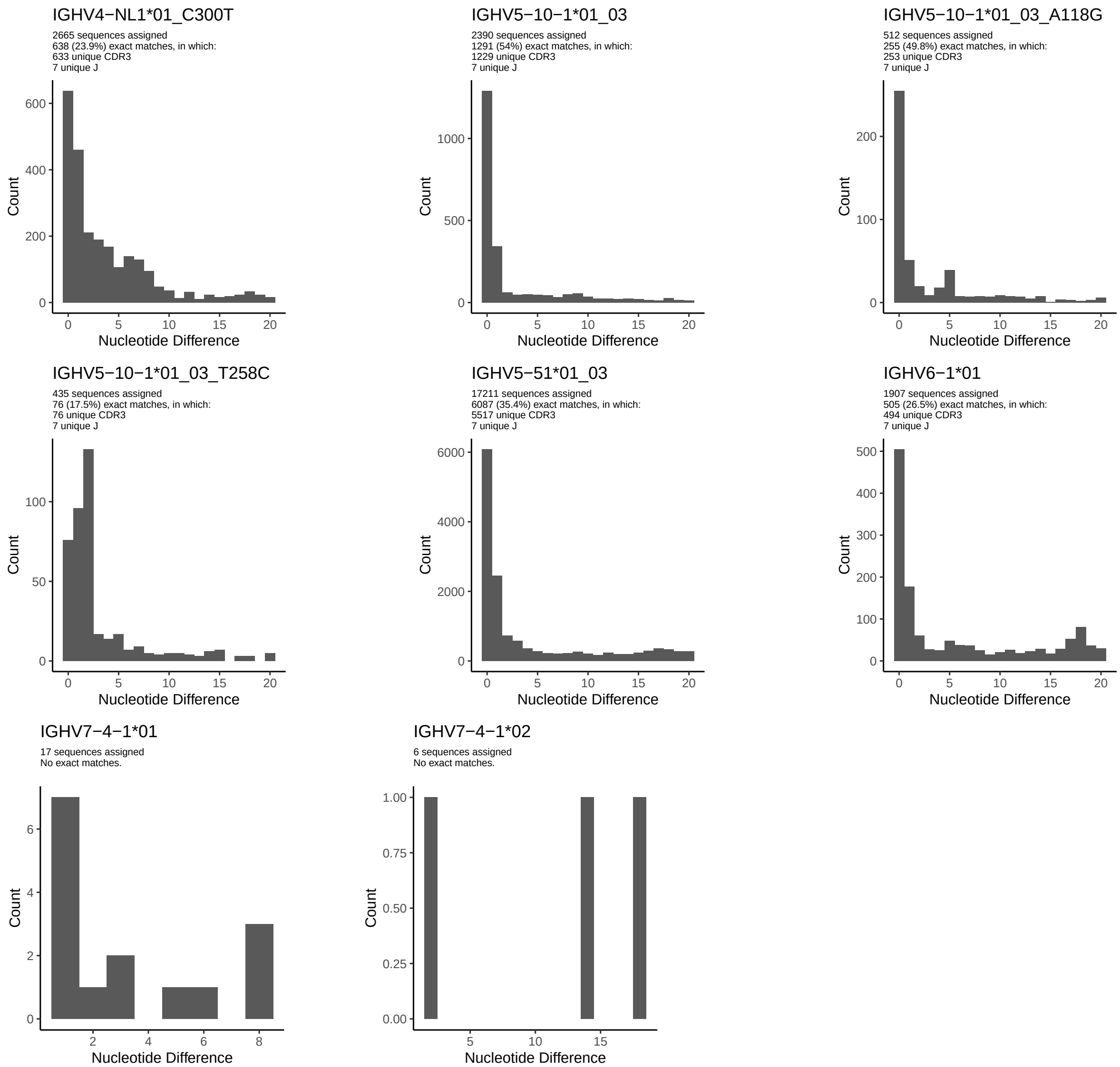

IGHV3−23*01_04
30336 sequences assigned
8892 (29.3%) exact matches, in which:
8376 unique CDR3
7 unique J
IGHV3−23*01_04_A303G_A317G
10930 sequences assigned
440 (4%) exact matches, in which:
431 unique CDR3
7 unique J
IGHV3−25*04
1 sequences assigned
No exact matches.
IGHV3−30−3*01
3237 sequences assigned
1399 (43.2%) exact matches, in which:
1367 unique CDR3
7 unique J
IGHV3−30*03
1174 sequences assigned
347 (29.6%) exact matches, in which:
347 unique CDR3
7 unique J
IGHV3−30*18
4228 sequences assigned
1449 (34.3%) exact matches, in which:
1410 unique CDR3
7 unique J
IGHV3−30−3*01_T288C
943 sequences assigned
151 (16%) exact matches, in which:
151 unique CDR3
7 unique J
IGHV3−30−3*01_T288C_T300C_G
650 sequences assigned
210 (32.3%) exact matches, in which:
210 unique CDR3
7 unique J
IGHV3−30*03_T288C
1371 sequences assigned
291 (21.2%) exact matches, in which:
291 unique CDR3
7 unique J
IGHV3−30*18_T288C_T300C_G30
1041 sequences assigned
273 (26.2%) exact matches, in which:
272 unique CDR3
7 unique J
IGHV3−33*01
3715 sequences assigned
1522 (41%) exact matches, in which:
1491 unique CDR3
7 unique J
IGHV3−33*08
1390 sequences assigned
291 (20.9%) exact matches, in which:
291 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV3−33*06_C288T
565 sequences assigned
232 (41.1%) exact matches, in which:
231 unique CDR3
7 unique J
IGHV3−33*06_T300C_G303A
892 sequences assigned
226 (25.3%) exact matches, in which:
226 unique CDR3
7 unique J
IGHV3−35*01
7 sequences assigned
3 (42.9%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV3−43*01
514 sequences assigned
149 (29%) exact matches, in which:
144 unique CDR3
7 unique J
IGHV3−48*01
792 sequences assigned
123 (15.5%) exact matches, in which:
123 unique CDR3
7 unique J
IGHV3−48*02
5124 sequences assigned
1559 (30.4%) exact matches, in which:
1509 unique CDR3
7 unique J
IGHV3−48*04
886 sequences assigned
108 (12.2%) exact matches, in which:
108 unique CDR3
6 unique J
IGHV3−48*01_T300C
931 sequences assigned
56 (6%) exact matches, in which:
56 unique CDR3
6 unique J
IGHV3−49*03_05
812 sequences assigned
275 (33.9%) exact matches, in which:
267 unique CDR3
7 unique J
IGHV3−53*01_02
5088 sequences assigned
1067 (21%) exact matches, in which:
1032 unique CDR3
7 unique J
IGHV3−64*01
375 sequences assigned
118 (31.5%) exact matches, in which:
113 unique CDR3
7 unique J
IGHV3−64*02_07
145 sequences assigned
16 (11%) exact matches, in which:
16 unique CDR3
4 unique J
Count
Nucleotide Difference

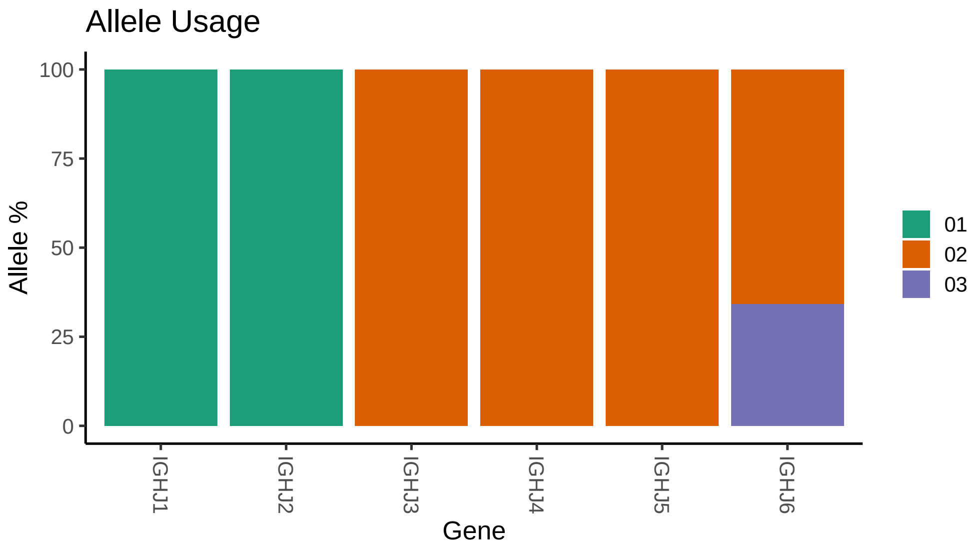

IGHV3−66*01
19 sequences assigned
1 (5.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−66*02
339 sequences assigned
No exact matches.
IGHV3−69−1*01
53 sequences assigned
8 (15.1%) exact matches, in which:
8 unique CDR3
6 unique J
IGHV3−69−1*02
51 sequences assigned
No exact matches.
IGHV3−72*01
813 sequences assigned
37 (4.6%) exact matches, in which:
37 unique CDR3
5 unique J
IGHV3−73*01_02
579 sequences assigned
193 (33.3%) exact matches, in which:
190 unique CDR3
7 unique J
IGHV3−74*01
3635 sequences assigned
549 (15.1%) exact matches, in which:
522 unique CDR3
7 unique J
IGHV3−74*01_A315G
2406 sequences assigned
112 (4.7%) exact matches, in which:
112 unique CDR3
7 unique J
IGHV3−43D*04_05
15 sequences assigned
No exact matches.
IGHV3−64D*06
1163 sequences assigned
375 (32.2%) exact matches, in which:
351 unique CDR3
7 unique J
IGHV3−NL1*01
75 sequences assigned
No exact matches.
IGHV4−4*02_03
1467 sequences assigned
623 (42.5%) exact matches, in which:
606 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−4*02_03_A245C_C300T
574 sequences assigned
101 (17.6%) exact matches, in which:
101 unique CDR3
7 unique J
IGHV4−28*03
3 sequences assigned
No exact matches.
IGHV4−28*01_07
11 sequences assigned
4 (36.4%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV4−30−2*01
771 sequences assigned
243 (31.5%) exact matches, in which:
231 unique CDR3
6 unique J
IGHV4−30−4*01
1195 sequences assigned
355 (29.7%) exact matches, in which:
349 unique CDR3
7 unique J
IGHV4−30−4*08
342 sequences assigned
3 (0.9%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV4−30−2*01_C315G
823 sequences assigned
61 (7.4%) exact matches, in which:
61 unique CDR3
6 unique J
IGHV4−30−4*01_C315G
1039 sequences assigned
72 (6.9%) exact matches, in which:
72 unique CDR3
7 unique J
IGHV4−31*03
6238 sequences assigned
1756 (28.2%) exact matches, in which:
1721 unique CDR3
7 unique J
IGHV4−34*01
15781 sequences assigned
5326 (33.7%) exact matches, in which:
5072 unique CDR3
7 unique J
IGHV4−34*01_C288T_T300C
2427 sequences assigned
916 (37.7%) exact matches, in which:
912 unique CDR3
7 unique J
IGHV4−34*01_G291A
2699 sequences assigned
572 (21.2%) exact matches, in which:
569 unique CDR3
7 unique J
Count
Nucleotide Difference

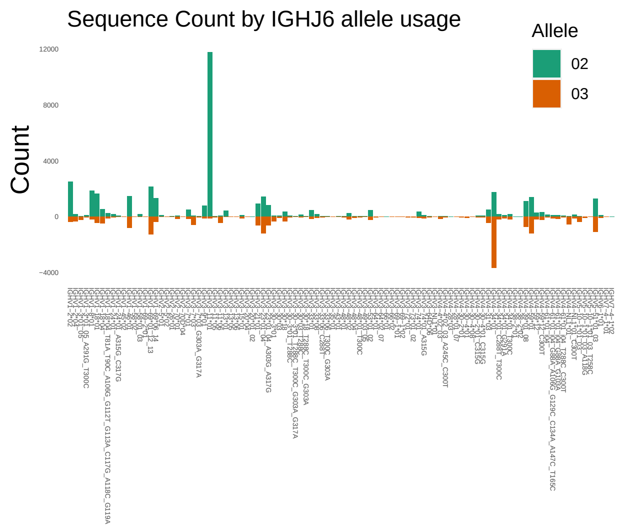

IGHV4−34*01_T300C
2619 sequences assigned
511 (19.5%) exact matches, in which:
508 unique CDR3
7 unique J
IGHV4−38−2*01
8 sequences assigned
No exact matches.
IGHV4−38−2*02
38 sequences assigned
No exact matches.
IGHV4−39*01_08
15594 sequences assigned
3691 (23.7%) exact matches, in which:
3555 unique CDR3
7 unique J
IGHV4−59*01
15952 sequences assigned
5609 (35.2%) exact matches, in which:
5402 unique CDR3
7 unique J
IGHV4−59*12
3513 sequences assigned
363 (10.3%) exact matches, in which:
361 unique CDR3
7 unique J
IGHV4−59*12_C300T
3052 sequences assigned
1149 (37.6%) exact matches, in which:
1140 unique CDR3
7 unique J
IGHV4−61*01_i04
1496 sequences assigned
658 (44%) exact matches, in which:
651 unique CDR3
7 unique J
IGHV4−61*01_i04_G88A_A106G_
1504 sequences assigned
170 (11.3%) exact matches, in which:
170 unique CDR3
7 unique J
IGHV4−61*01_i04_G88A_G103A
1561 sequences assigned
175 (11.2%) exact matches, in which:
174 unique CDR3
7 unique J
IGHV4−61*01_i04_T288C_C300T
712 sequences assigned
102 (14.3%) exact matches, in which:
102 unique CDR3
7 unique J
IGHV4−NL1*01
6301 sequences assigned
1742 (27.6%) exact matches, in which:
1689 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV4−NL1*01_C300T
2665 sequences assigned
638 (23.9%) exact matches, in which:
633 unique CDR3
7 unique J
IGHV5−10−1*01_03
2390 sequences assigned
1291 (54%) exact matches, in which:
1229 unique CDR3
7 unique J
IGHV5−10−1*01_03_A118G
512 sequences assigned
255 (49.8%) exact matches, in which:
253 unique CDR3
7 unique J
IGHV5−10−1*01_03_T258C
435 sequences assigned
76 (17.5%) exact matches, in which:
76 unique CDR3
7 unique J
IGHV5−51*01_03
17211 sequences assigned
6087 (35.4%) exact matches, in which:
5517 unique CDR3
7 unique J
IGHV6−1*01
1907 sequences assigned
505 (26.5%) exact matches, in which:
494 unique CDR3
7 unique J
IGHV7−4−1*01
17 sequences assigned
No exact matches.
IGHV7−4−1*02
6 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots



5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 086/M64 086/M64 086/M64 086/02/43e325438881e6108
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 086/M64 086/M64 086/M64 086/02/43e3254388
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 086/M64 086/M64 086/M64 086/02/43e325438881e610
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 18 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 18 04_T81A_T90C_A106G_G112T_G113A_C117G_A118C_G119A: replacing w
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 7 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 7 03_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 18: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C_T300C_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 34 01_G291A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 01_i04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_T288C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_G129C_C134A_A147C_T165C: replacing with ga
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV5 10 1 01_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV5 10 1 01_03_A118G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV5 10 1 01_03_T258C: replacing with gap
```